8 A cylinder is placed on a table.

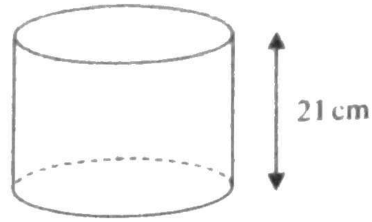

Diagram **NOT** accurately drawn

The volume of the cylinder is 1575 cm^3
The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

Volume of ~~Cy~~linder = $\pi r^2 h$

$$\frac{1575}{\pi\, 21} = \frac{\cancel{\pi} r^2 \cancel{h}\ \cancel{21}}{\cancel{\pi}\ \cancel{21}}$$

$$r^2 = \frac{1575}{\pi 21}$$

$$r = \sqrt{\frac{1575}{21\pi}}$$

$$r = 4.886$$

$\therefore$ Area of Cylinder = $2\pi r h + 2\pi r^2$

$= 2 \times \pi \times 4.886 \times 21 + 2 \times \pi \times 4.886^2$

$= 794.690$

$\therefore$ Pressure = $\dfrac{84}{794.690}$

$= 0.1057$

$= 0.106$ newton/cm^3

0.106 newtons/cm^2

(Total for Question 8 is 3 marks)

DO NOT WRITE IN THIS AREA

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